

# Technical Task — VLM Pipeline for Manufacturing-Drawing Part Classification

One drawing → part type (sheet / tube) → bend count, with view / title-block region localization.

AAI Labs | 2026

## 1 Objective & scope

Ingest one single-part manufacturing PDF drawing and return

- part class **sheet** or **tube**;
- for sheets, the **number of bends**;
- bounding boxes for each view and the title/BOM block, cropped and fed to (a) and (b) so each step works on a clean sub-image.

## 2 Pipeline — three sequential VLM stages

| Stage         | Input                                   | VLM task & key cues                                                                                                                                                                                                                                                                               | Output                |
|---------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1 Localize    | full-page raster                        | Grounding VLM → labeled boxes for region types <b>flat_pattern</b> , <b>orthographic_view</b> , <b>isometric_view</b> , <b>section_view</b> , <b>title_block</b> . Prompt for JSON {label, box_2d}; Qwen-family expects <b>XYXY normalized 0–1000</b> (coordinate order/scale affects accuracy).  | boxes + crops         |
| 2 Classify    | title_block + view crops                | <b>sheet vs. tube</b> . Sheet cues: Abwicklung / Blechdicke, thin uniform edge view, desc PLATE·BRACKET·BLECH. Tube cues: constant hollow section + long length, section view with wall thickness, desc ...TUBE·ROHR·PROFIL.                                                                      | class, conf, evidence |
| 3 Count bends | flat_pattern + side/section + isometric | Bend = one fold line where the plane changes direction = (straight planar segments in thinnest folded profile – 1); corroborate with flat-pattern fold lines + isometric. <b>Exclude</b> chamfers (n×45°), edge fillets/radii, holes, stepped-outline cuts. Radius callouts (R2 4x) ≠ bend count. | num_bends, evidence   |

## 3 Reference samples (ground truth for the eval set)

| Drawing | Description          | Material / t  | Class | Bends | Discriminating signal                                |
|---------|----------------------|---------------|-------|-------|------------------------------------------------------|
| 001     | BRACKET              | 1.4301 / 3 mm | sheet | 2     | Abwicklung present; U-channel side profile = 2 folds |
| 002     | PLATE                | AlMg3 / 3 mm  | sheet | 0     | Flat plate; corner chamfers 2×45° are NOT bends      |
| 003     | SQUARE TUBE 80×80×10 | S235 / —      | tube  | n/a   | Hollow constant section + 2790 mm length             |
| 004     | ADAPTER PLATE        | S235 / 10 mm  | sheet | 0     | Flat bar; stepped outline is a cut, NOT a fold       |

## 4 Suggested output schema & evaluation

```
{ "drawing_id": "TSL-12449-593", "regions": [
  { "type": "flat_pattern", "bbox": [x0,y0,x1,y1], "conf": 0.94 }, ... ], // types:
  flat_pattern|orthographic_view|isometric_view|section_view|title_block
  "class": "sheet", "class_confidence": 0.98, "class_evidence": "Abwicklung + Blechdicke
  3mm; desc BRACKET", "num_bends": 2, "bend_confidence": 0.90, "bend_evidence": "side
  profile = 3 segments -> 2 folds", "flags": [] }
```

- **Localization**: correctly localize projection views and BOM tables
- **Classification**: precisely discriminate TUBE vs SHEET
- **Bend count**: correctly count number of bends for the SHEET
- **End-to-end**: pipeline works end-to-end, where the input is a raw PDF drawing

## 5 Desired Output and Deliverables

- **README.md**
- **Pipeline code** — the three-stage runner (localize → classify → count) with prompts, configs and crop logic versioned; deterministic where possible (fixed DPI, temperature 0, seeds). Batch mode over a folder of PDFs.
- **Model-choice rationale** — which VLM(s) were selected for each stage and why, backed by the \$4 eval (accuracy vs. latency vs. cost): the alternatives benchmarked and the trade-off that decided it — open-weight/self-hosted vs. API.
- **Inference-time & hardware analysis** — per-stage and end-to-end latency, throughput (drawings/min), peak VRAM, and the exact GPU / precision (fp16 / int8) used; note whether Stage-1 grounding or Stage-3 reasoning dominates. State the measurement hardware so the numbers are comparable.
  - For benchmarking without buying hardware, rent on-demand GPUs — e.g. **vast.ai** (cheap marketplace GPUs) and record the instance type (e.g. RTX 4090 / A100) next to the timings.
- **Sample outputs** — emitted JSON and annotated crops for the 4 reference drawings, so results are visually checkable.